

Max Matching Characterization

From last lecture:

A matching M in a graph $G=(V,E)$
is maximum iff there does not exist
an M -aug path.

Holds for general (not only bipartite)
graphs.

Tutte-Berge Theorem

$$G = (V, E)$$

$\nu(G)$ = size of maximum matching

$o(G)$ = # of odd components of a graph

Thm: $\nu(G) = \min_{U \subseteq V} \frac{1}{2} (|V| + |U| - o(G-U))$

Corollary for M any matching, if $\exists \hat{U} \subseteq V$ s.t.
 $|M| = \frac{1}{2} (|V| + |\hat{U}| - o(G-\hat{U})) \Rightarrow M$ is max size.

Corollary G has a perfect matching iff $\forall U \subseteq V, o(G-U) \leq |U|$.
pf: plug back into result, with $\nu(G) = \frac{1}{2}|V|$.

Proof of Tutte-Berge — 1/4

" $\leq$ "

$$\forall u \subseteq V$$

$$\nu(G) \leq |u| + \nu(G-u)$$

$$\leq |u| + \frac{1}{2} (|V \setminus u| - o(G-u))$$

$$= \frac{1}{2} (|V| + |u| - o(G-u))$$

Proof of Tutte-Berge — 2/4

" $\geq$ " By induction on $|V|$

$|V| = 0$ or 1 immediate.

why G is connected.

Suppose, $\exists$ vertex $\hat{v}$ covered by all
max size matchings.

$$\Rightarrow v(G - \hat{v}) = v(G) - 1.$$

$\uparrow$
can apply inductive hyp here and conclude result.

$$v(G - \hat{v}) = \frac{1}{2} (|V| - 1 + |U'| - o(G - U' - \hat{v}))$$

$$\text{Take } U = U' \cup \{\hat{v}\}$$

$$\Rightarrow v(G) = v(G - \hat{v}) + 1 = \frac{1}{2} (|V| + |U| - o(G - U)) .$$

Proof of Tutte-Berge — 3/4

Thus can assure $\forall v$ covered by all ^{max} matchings

$$\Rightarrow v(G) < \frac{1}{2}|V|$$

We will show $\exists M$ with $|M| = \frac{1}{2}(|V| - 1)$.

To obtain a contradiction:

Assume any max M has $|M| < \frac{1}{2}(|V| - 1)$

i.e., any M misses at least 2 nodes: u, v

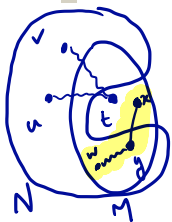
Choose: (M, u, v) among max size matchings s.t.

so that $\text{dist}(u, v)$ is minimal.

Certainly: $\text{dist}(u, v) \geq 2$.

$u \dots t \dots v$

let $t \neq u, v$ be a node on the shortest $u-v$.



Since $\text{dist}(u, v)$ is minimal, u, v covered by N .

N misses t , (covers u, v), must also fail to cover some other node: x .

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$\exists y$ with $(x, y) \in M$, i.e., M covers x .

Claim: $\exists (y, w) \in N$, i.e., y is covered by N .

Now: $(N \setminus \{y, w\}) \cup \{y, x\}$ is a max matching that misses t
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